Experiment 8

Question

Scaling an image to its Bigger and Smaller sizes.

Aim

To read an image using Python and OpenCV and scale it to bigger and smaller sizes.

Procedure

1. Import the OpenCV (cv2) library.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Display the original image.
5. Resize the image to a larger size.
6. Resize the image to a smaller size.
7. Display both the enlarged and reduced images.
8. Save the output images.

Code

```
import cv2
# Read image
image = cv2.imread("penguin.jpg")

# Check whether image is loaded
if image is None:
   print("Image not found!")
else:
   # Display original image
   cv2.imshow("Original Image", image)

   # Scale image to bigger size
   bigger = cv2.resize(image, (800, 600))

   # Scale image to smaller size
   smaller = cv2.resize(image, (300, 200))

   # Display bigger image
   cv2.imshow("Bigger Image", bigger)

   # Display smaller image
   cv2.imshow("Smaller Image", smaller)
```

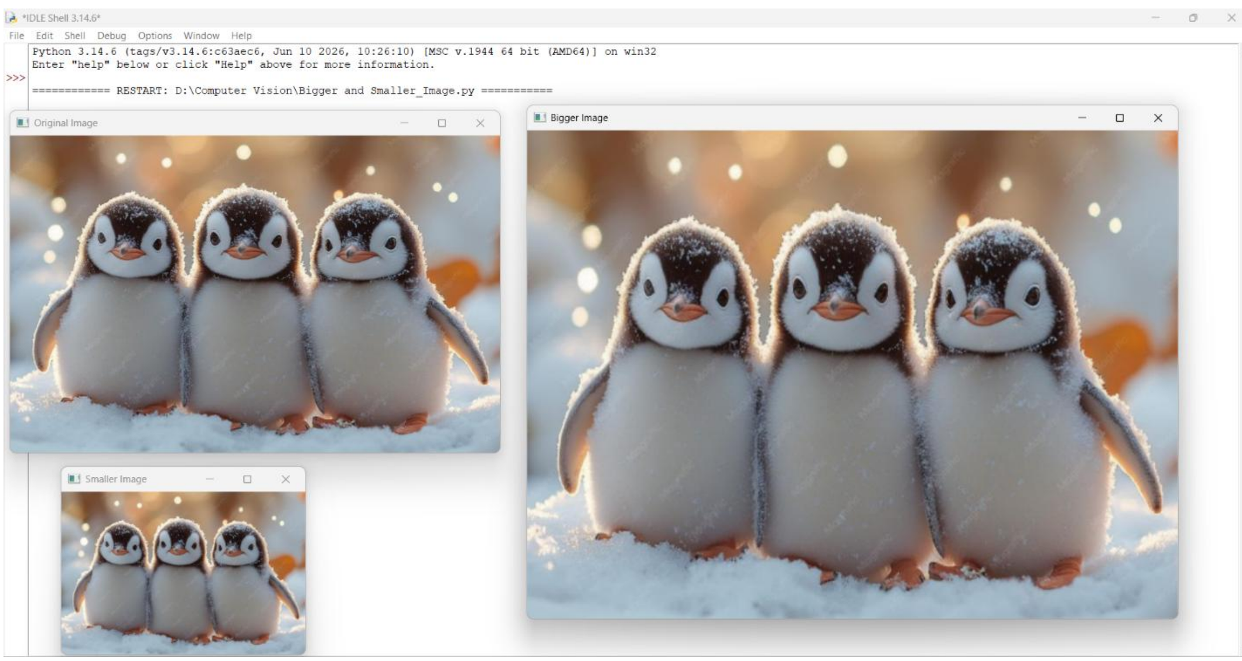
```
# Save output images
cv2.imwrite("bigger_penguin.jpg", bigger)
cv2.imwrite("smaller_penguin.jpg", smaller)
```

Input

Input Image: penguin.jpg

Output

1. **Original Image**
2. **Bigger Image (800 × 600)**
3. **Smaller Image (300 × 200)**



Result

The image was successfully read using OpenCV and resized to both bigger and smaller dimensions. The resized images were displayed and saved successfully.